SAFETY ENGINEERING SINCE 2004

Industrial & Environmental Risk Analysis

We turn technical complexity into safe, reliable and compliant decisions: process risk, environment, reliability and human factors.

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OUR STORY

A company born from research

20+ Years of experience	**500+** Risk studies and analyses
7 Areas of expertise	**2005** ISO 9001 certified since

ARIA was founded in 2004 and, since **2007, has been a graduate company of I3P – the Innovative Business Incubator of the Politecnico di Torino**. For over twenty years we have developed and applied innovative methodologies for industrial and environmental risk analysis, reliability and human factors.

Our work combines scientific rigour with field experience, fuelled by ongoing participation in **national and international research projects** that keep our expertise at the state of the art.

- ✓ **ISO 9001** certification issued by DNV since 2005
- ✓ Proprietary risk-analysis methodologies
- ✓ Multidisciplinary approach: process, environment, human factors

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> **We combine independence of judgement and methodological rigour to deliver clear, defensible assessments truly oriented to decision-making.**
>
> — The ARIA method

OUR VALUES

How we work

Four principles guide every project.

Scientific rigour

Validated, traceable methods grounded in academic research and sector regulations.

Methodological innovation

We develop and apply new risk-analysis techniques born from research projects.

Multidisciplinary approach

We integrate process, environment and the human factor into a unified view of risk.

Independence

Third-party, impartial assessments dedicated solely to the client's safety.

THE JOURNEY

Key milestones

2004

ARIA is founded, out of the research environment of the Politecnico di Torino.

2005

Achievement of ISO 9001 certification issued by DNV.

2007

ARIA becomes a graduate company of I3P – the Innovative Business Incubator of the Politecnico di Torino.

Today

Ongoing participation in national and international research projects and consulting for the main industrial sectors.

WHAT WE DO

Seven areas of expertise, one goal: safety

Innovative risk-analysis methodologies applied to the process industry, the environment and human factors.

01 Industrial safety

Process risk: from hazard identification to decision support.

02 Workplaces

Occupational health and safety (Legislative Decree 81/08).

03 Reliability and maintainability

RAMS analysis, SIL and maintenance optimisation.

04 Product safety

CE marking, compliance and the technical file.

05 Management systems

Quality, environment, safety and energy, risk-based.

06 Training

Tailored programmes for every corporate level.

07 Human factors

Risk analysis integrated with human and organisational factors.

01

Industrial safety

Risk analysis of complex technological systems: from hazard identification to decision support, using established and innovative methodologies.

Hazard and risk identification

- **HazOp** – Hazard and Operability Studies, using classic techniques (guide words) and advanced techniques (Recursive Operability Analysis)
- **FMECA** – Failure Modes and Effect Analysis
- Safety Audits
- Preliminary Hazard Analysis (PHA)
- Hazard Identification Analysis (HazId)

Quantification of occurrence probabilities

- Fault trees (**Fault Tree Analysis – FTA**)
- Event trees (**Event Tree Analysis – ETA**)
- Safety Integrity Level (SIL)
- Integrated Dynamic Decision Analysis (IDDA)

Consequence assessment

- Mathematical modelling of explosions, dispersions and fires
- Consequences of **BLEVE, UVCE, CVE** and thermal radiation (pool and jet fires)

Risk representation and decision support

- Layer Of Protection Analysis (LOPA)
- Bow-Tie Analysis
- Integrated Dynamic Decision Analysis (IDDA)

Land-use planning

- Land-use planning for companies (Italian Ministerial Decree of Public Works of 9 May 2001)
- Support to municipalities: environmental and land-use compatibility of major-accident activities (RIR Technical Report)

Operational experience analysis

- Analysis of accidents, near misses and anomalies in support of risk assessment
- Analysis of unsafe behaviours and conditions for preventive purposes

02

Workplaces

Occupational health and safety in the workplace (Italian Legislative Decree 81/08 as amended): we identify and apply the regulatory obligations that protect people.

General and specific risk assessment

- REACH and chemical risk
- ATEX (explosive atmospheres)
- Noise and vibration
- Electrical risk and biological risk
- Manual handling of loads
- Exposure to carcinogens and to display screen equipment

Fire safety

Fire risk assessment and preparation of the documentation required for the Fire Brigade to issue the Fire Prevention Certificate (CPI).

Emergency planning

Drafting of the Internal Emergency Plan (PEI) and definition of emergency procedures.

Analysis of incidents and accidents

Analysis of incidents, near misses and accidents using innovative analysis techniques (Fuzzy Logic).

Management systems (OHS)

Development and integration of Occupational Health and Safety Management Systems (ISO 45001).

Information and training

Information, training and instruction of workers on risks and on prevention measures.

Machinery Directive and Construction Sites Directive

- Machinery Directive: risk assessment and support for CE/ATEX certification
- Construction Sites Directive

03

Reliability and maintainability

RAMS analysis of complex plants and technological systems, using traditional techniques and innovative approaches to maximise performance and safety throughout the life cycle.

RAMS analysis

Assessment of reliability and availability using traditional techniques (FMECA, FTA) and innovative ones (Markov chains, IDDA, Fuzzy Logic). The same techniques support maintenance optimisation (Reliability Centred Maintenance – RCM).

SIL assessment

Safety Integrity Level of process plants (ISO 61511), operating machinery (ISO 62061) and the automotive sector (ISO 26262).

Reliability Databases

Built on company-specific information or with reference to the commercial databases of leading bodies (ESReDA, OREDA, AIChE, et al.).

Maintenance data collection systems

Design of systems for collecting, storing and analysing plant maintenance data, supporting the development of reliability databases.

Integrated Dynamic Decision Analysis (IDDA)

Combines logical-probabilistic modelling (a "dynamic" event tree) with phenomenological modelling of the system's physical response, for documented, justifiable decision support with an analysis of the alternatives and the risk involved.

04

Product safety

We support the product towards compliance: from legislative and regulatory compliance and CE-marking support to risk assessment and the drafting of the technical file.

Regulatory scouting and CE marking

We identify the directives and regulations applicable to the product according to its target markets — CE marking for Europe, FDA or UL requirements for the United States — defining the project's legal compass before investing in production.

Chemical and environmental compliance

We verify compliance with directives on hazardous substances and environmental impact — RoHS, REACH, SCIP and the packaging regulation — for products that remain compliant across the whole supply chain.

Risk assessment

We apply standard methodologies — FMEA (Failure Mode and Effects Analysis) and ISO 12100 for machinery — to identify mechanical, electrical, thermal and chemical hazards and correct them in advance.

Technical file and documentation

We draft the Technical File (drawings, calculations, test reports), user manuals, warnings and hazard labels, and support the preparation of the Declaration of Conformity through which the manufacturer assumes legal responsibility for the product.

05

Management systems

Management systems for quality, environment, safety and workers' health, built with a risk-based approach and integrated into business processes.

ISO 9001 • Quality

ISO 14001 • Environment

ISO 45001 • Safety

UNI 10617/10616 • Safety management

ISO 50001 • Energy

Initial analysis

Initial analysis of safety and/or environmental obligations, to define the status quo from which to begin developing the management system.

Operating experience analysis

Analysis of accidents, near misses and anomalies to support risk assessment and the optimisation of production and maintenance processes.

Management system development

Drafting of the documents that form the backbone of the system (manual, procedures and operating instructions), in close collaboration with the client company.

Audits

Interim audits to verify the state of implementation, pre-audits for certification purposes, and participation in audits by the certification body and/or supervisory authorities.

Training and instruction

Specific training for the management team, the operational structure, internal auditors and staff.

Risk-based approach and integration

Support in introducing the risk-based approach (ISO 9001 and ISO 14001, LCP and LCA) and integration of management systems to optimise company resources.

06

Training

Worker information, education and on-site training: tailored programmes to transfer skills and a culture of safety at every corporate level.

Courses for all corporate levels

- Employers
- Managers of major-accident hazard plants
- Supervisors and workers
- Workers' Safety Representatives (RLS)
- Emergency response and first-aid personnel

In-depth courses on specific risks

- Chemical risk and fire risk
- Explosive atmospheres (ATEX)
- Machinery Directive and product certification
- Risk-analysis techniques
- Safety Integrity Level (SIL)

Product safety

Courses on CE marking, product directives and regulations: risk assessment (ISO 12100), technical file and declaration of conformity.

Environmental training

Waste management, atmospheric emissions and the transport of dangerous goods (ADR Agreement).

Participatory safety

Collaborative programmes for groups of workers that integrate risk perception into assessments.

Qualified instructors and funding

Trainers holding the qualification requirements of the Italian Interministerial Decree of 6 March 2013 (art. 6(8)(m-bis), Legislative Decree 81/2008). Support in designing tailored programmes and accessing funding (e.g. Fondimpresa).

07

Human factors

Technological risk analysis integrated with human and organisational factors, and the optimisation of operating procedures for safety.

Procedure analysis and optimisation

Development of safety-oriented operating procedures, creating the conditions for operators and work teams to contribute actively to safety.

Integrated risk assessment

Assessment of technological process risk integrated with the analysis of human and organisational factors, extended also to workplaces.

Operational-experience tools

Development of tools for collecting, analysing and using plant operational experience: unsafe acts and conditions, and the transfer of information between shifts (shift handovers).

PROJECTS AND RESEARCH

Research projects and collaborations

We take part in national and international research projects on safety, risk analysis, the environment and human factors.

ASSOCIATED PARTNER · 2021–2025

CISC

Marie Skłodowska-Curie ITN · Horizon 2020 (EU)

Collaborative Intelligence for Safety Critical Systems: an international training network for researchers at the crossroads of artificial intelligence, human factors and systems safety engineering. ARIA contributed as associated partner.

LEAD PARTNER · 2020–2023

SALS

Finpiemonte · POR-FESR 2014/2020 – Innovation Clusters

Food-safety management along the supply chain with advanced tools: innovative sensors and real-time data-sharing networks to extend control across the entire chain.

PARTNER · 2020–2021

SAFELIVERY

EIT-FOOD · Crisis Response Initiative (EU)

A safer food-delivery service during and after the COVID-19 pandemic. ARIA developed HACCP procedures integrated with human factors and a prototype tamper-proof box with a smart lock.

PARTNER · 2011–2016

INNHF

Innovation through Human Factors in Risk Analysis and Management

Integration of human and managerial factors into traditional risk-analysis methodologies. ARIA hosted a researcher for three years, focused on human factors in ATEX procedures.

PARTNER · 2014–2015

GEOMAT

Regione Piemonte · POR FESR 07/13

Geopolymer binders as an alternative to Portland cement. ARIA carried out preliminary Life Cycle Analysis studies for the technical and economic assessment of the product.

PARTNER · 2012–2014

DUALCEM

Regione Piemonte · POR FESR 07/13

High-durability self-healing cementitious materials without compromising their structural properties. ARIA provided analysis in support of the development work and a preliminary Life Cycle Analysis.

SECTORS

Where we operate

We support organisations in the sectors most exposed to industrial and environmental risk.

- **Chemical & Petrochemical**
- **Oil & Gas**
- **Energy**
- **Pharmaceutical**
- **Manufacturing**
- **Infrastructure**
- **Environment & Waste**
- **Transport**

SCIENTIFIC PUBLICATIONS

Our contributions to research

Articles in international journals and conference proceedings on risk analysis, safety, human factors and land use planning. A selection:

- Geng J.; Murè S.; Demichela M.; Baldissone G. (2020). **Atex/HOF methodology: Innovation driven by human and organizational factors in explosive atmosphere risk assessment**. *Safety* 6(1), 5.
- Baldissone G.; Comberti L.; Bosca S.; Murè S. (2019). **The analysis and management of unsafe acts and unsafe conditions**. *Safety Science* 119, pp. 240-251.
- Pilone E.; Demichela M.; Baldissone G. (2019). **The multi-risk assessment approach as a basis for territorial resilience**. *Sustainability* 11(9), 2612.
- Demichela M.; Baldissone G.; Camuncoli G. (2017). **Risk-Based Decision Making for the Management of Change in Process Plants**. *Industrial and Engineering Chemistry Research* 56(50), pp. 14873-14887.
- Murè S.; Comberti L.; Demichela M. (2017). **How harsh work environments affect occupational accident phenomenology?** *Safety Science* 95, pp. 159-170.
- Demichela M.; Camuncoli G. (2014). **Risk based decision making. Discussion on two methodological milestones**. *Journal of Loss Prevention in the Process Industries*, pp. 101-108.
- Pilone E.; Mussini P.; Demichela M.; Camuncoli G. (2016). **Municipal Emergency Plans in Italy: requirements and drawbacks**. *Safety Science*, pp. 163-170.
- Murè S.; Demichela M. (2009). **Fuzzy Application Procedure (FAP) for the risk assessment of occupational accidents**. *Journal of Loss Prevention in the Process Industries*, pp. 593-599.

Full list (over 40 publications from 2006 onward) at **www.aria.srl/en/publications**

WHAT'S NEW

Packaging Vademecum

The operational tool developed by ARIA and ICIM Consulting for compliance in the management of packaging waste across Europe. Pending a European Regulation harmonising the application of EU Directive 2018/852, the Vademecum describes, through simple and intuitive navigation, the obligations to be met in each European country: packaging labelling requirements, membership of eco-organisations and declarations of the quantities of packaging placed on each market.

environmentallabellingpackaging.com

CLIENTS

Companies that have placed their trust in us

Leading industrial players and institutions in the chemical, pharmaceutical, energy, mechanical and food sectors rely on ARIA for risk analysis and management.

ARIA Srl

Industrial and Environmental Risk Analysis

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